SMART INDIA HACKATHON 2026

SMART INDIA HACKATHON 2026

TITLE PAGE

Problem Statement ID – **SIH26011**

Problem Statement Title – 3D ULPIN (Bhu-Aadhaar) Generation, Vertical Cadastre & 3D Building Visualization Digital Twin

Theme – Smart Automation / Geospatial Technology / Smart Cities

PS Category – Software

Team Name (Registered on portal) – **GeoVoxel The coders**

GeoVoxel
The coders

GEO-CADASTRE 3D: Autonomous 3D ULPIN & Vertical Cadastre Digital Twin

SMART INDIA HACKATHON 2026

Proposed Solution

Autonomous geospatial pipeline generating India's first 3D vertical cadastre from open-source satellite and survey data.

60 FPS WebGL digital twin with real-time building inspection, floor isolation, and solar analysis.

Technical Approach

Tier 1 — Unit-Level: Parses RERA blueprints into individual flat parcels with carpet area, facing, and Z-coordinates.

Tier 2 — Floor-Level: Automated slicing via DSM-DEM height estimation with zero-discrepancy normalization.

Problem Addressed

Replaces ambiguous 2D survey numbers with unique 3D spatial identities for every apartment unit.

ISO 19152 LADM + OGC CityGML compliant legal 3D parcels (Z-min, Z-max MSL).

Innovation

Zero-Discrepancy Floor Slab Engine: cumulative height error = 0.00m across 4,641 buildings.

Client-side 60 FPS rendering of 22,761 volumetric parcels — no server GPU required.

GOVERNMENT 2D BHU-AADHAAR (ULPIN)

3621050103VUWK 14-Digit DoLR Standard

State 36 | District 21 | Mandal 05 | Village 0103 | Survey 83/1

TIER 1: UNIT 3D ULPIN

Sanctioned RERA Blueprints

- **FL32-U3201**
  4BHK Luxury | 318 m² | East
- **FL32-U3202**
  3BHK Premium | 241 m² | West
- **FL32-U3203**
  3BHK Premium | 241 m² | North
- **FL32-ULOBBY**
  Common Area | Lift & Fire | 85 m²

TIER 2: FLOOR 3D ULPIN

Automated Height Slicing

- **FL32**
  Floor 32 | Z: 691.01 – 694.00m MSL
- **FL15**
  Floor 15 | Z: 639.75 – 642.75m MSL
- **FL08**
  Floor 8 | Z: 618.34 – 621.34m MSL
- **FL00**
  Ground | Z: 594.00 – 598.34m MSL

GeoVoxel
The coders

TECHNICAL APPROACH & PIPELINE ARCHITECTURE

SMART INDIA HACKATHON 2026

STAGE 1: DATA INGESTION

OpenStreetMap (Overpass QL)
Copernicus GLO-30 DSM
SRTM GL1 DEM (OpenTopo)
ISRO Bhuvan OGC Geoportal
TS-RERA & Dharani Land DB

STAGE 2: HEIGHT ENGINE

Landmark / RERA Registry
DSM-DEM Zonal Statistics
Annular Buffer Sampling
Morphological Typology
Cluster Spatial Propagation

STAGE 3: 3D CADASTRE

14-Digit Bhu-Aadhaar (NIC)
Unit ULPIN (-FLxx-Uyyyy)
Floor ULPIN (-FLxx)
Zero-Discrepancy Normalizer
ISO 19152 / CityGML 3.0

STAGE 4: DIGITAL TWIN

Next.js 16 + Three.js r185
60 FPS BatchedMesh / LOD
Interactive Floor Isolation
Solar & Flood Simulation
238 km Road Network

4,641
Buildings

22,761
3D Parcels

238 km
Roads

12.5s
Execution

60 FPS
Rendering

Technologies & Frameworks

Geospatial: Python 3.10+, GDAL, Rasterio, Shapely, GeoPandas, PyProj, SciPy k-d trees.

Remote Sensing: Copernicus GLO-30 DSM, SRTM GL1 DEM, ISRO Bhuvan WFS/WMS, OSM Overpass QL.

3D Platform: Next.js 16 (Turbopack), React 19, Three.js r185, React Three Fiber, WebGL 2.0.

Methodology & Implementation

1. **Ingestion:** Bounding box clipping, UTM reprojection, OSM vector extraction, Bhuvan validation.
2. **Height:** DSM-DEM zonal diff + annular buffer terrain sampling + typology heuristics.
3. **Cadastre:** 14-digit ULPIN, zero-discrepancy floor normalization, dual-tier volumetric parcels.
4. **Digital Twin:** Instanced rendering of 4,641 buildings + 238 km roads; interactive inspection.

GeoVoxel
The coders

FEASIBILITY AND VIABILITY ANALYSIS

SMART INDIA HACKATHON 2026

Feasibility Analysis

Data: 100% open sovereign data — Copernicus DEM, SRTM, ISRO Bhuvan, OSM. Zero licensing fees.

Operations: Entire 9 km² urban sector processed in 12.5 seconds on commodity hardware.

Technical: Built on ISO 19152 LADM, OGC CityGML, 14-digit DoLR ULPIN standards.

Deployment: Client-side WebGL at 60 FPS in any browser — no GPUs or native installs.

Challenges & Risks

Raster Noise: 30m DSM/DEM data has height distortion in dense high-rise clusters.

Missing Plans: 95%+ of municipal buildings lack digitized floor plans for unit mapping.

Browser Limits: Simultaneous volumetric rendering can exhaust WebGL memory.

Elevation Drift: Rounding errors cause cumulative drift between slices and rooftops.

Mitigation Strategies

5-Tier Fallback: Annular terrain buffering + morphological typology + landmark registries.

Dual-Mode: Floor-level slicing as default; upgrades to unit-level when blueprints exist.

On-Demand LOD: Only selected building is volumetrically extruded; others stay as batched mesh.

Normalizer: Enforces cumulative height = building height with 0.00m error tolerance.

GeoVoxel
The coders

IMPACT AND MULTI-SECTOR BENEFITS

SMART INDIA HACKATHON 2026

Potential Impact on Target Audience

Dept. of Land Resources & Survey of India

Blueprint to transition 2D Bhu-Aadhaar into a national 3D cadastral framework.

Urban Local Bodies (GHMC / Smart City)

15–25% higher property tax via detection of unassessed floors and FAR violations.

Homebuyers & Citizens

Undisputed legal title to specific vertical airspace, replacing UDS agreements.

Banks & Mortgage Lenders

Instant 3D collateral verification; eliminates fraudulent multi-mortgaging.

Benefits (Social, Economic, Environmental)

Economic

Eliminates land litigation costs; accelerates bank loan approvals from weeks to minutes.

Social & Governance

100% transparency in vertical ownership; prevents double-selling scams.

Disaster Management

Floor-level flood simulation identifies submerged levels for precision rescue.

Environmental

3D solar shadow analysis enables rooftop PV capacity estimation and heat island mitigation.

GeoVoxel
The coders

RESEARCH, REFERENCES & WORKING PROTOTYPE

SMART INDIA HACKATHON 2026

Research & Reference Work

DoLR / NIC Guidelines: ULPIN / Bhu-Aadhaar Technical Specifications & Design Guidelines (2021–2024).

ISO 19152 (LADM): Land Administration Domain Model — Part 3: 3D/4D Spatial Administration.

OGC Standards: CityGML 3.0 & 3D Tiles Community Standard for streaming geospatial models.

Survey of India & MoHUA: National Geospatial Policy 2022 & NUDM Smart City guidelines.

Academic Research: Biljecki et al. (2015): 3D City Models; Stoter et al. (2020): 3D Cadastre in Practice.

Data Sources: Copernicus GLO-30 DSM, SRTM DEM, ISRO Bhuvan, OSM (ODbL), TS-RERA Portal.

Working Prototype & Repository

GitHub: github.com/Imad-81/SIH-26011
3D Viewer: localhost:3000 (Next.js 16 + Three.js)
Pipeline: python scripts/pipeline.py --city hyderabad